

vision and hearing issues
safety and injury
sports injuries
pedestrian and bike safety
latchkey concerns

Needs of schools

A sense of achievement
Peer Acceptance
clear rules
Fair play
Encouragement not just praise

Get education

Get education

Role of parents

Supporting industry

Transitioning to the consultant

Modeling social and moral behavior

Monitoring the digital and social frontiers

Supporting the biological health

Role of Nurse

As a clinical screeners

As a Nutrition and activity advocate

As a emotional and social gatekeeper

As a Anticipatory guidance expert

Summary

Conclusion

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psycho social development

sense of industry vs inferiority

psycho sexual development

Latency stage

spiritual development

mythical vs literal faith

Intellectual development

concrete operational thought
Formal operational thought

Moral development

conventional morality

Language development

20,000 - 30,000 words

play

Social or competitive play

Nutrition

calories : 1600 - 2600 K cal / kg / day

protein : 3 - 4 gms / kg / day

Fat : 4 - 5 gm / kg / day

Iron : 15 - 20 µg / day

Vitamin A : 5000 IU / day

Common problems

Psychosocial and behavioral problems

school phobia / refusal

Bullying

Stress and overscheduling

physical and biological problems

Obesity

Dental caries

Add few

Gastrointestinal system

The stomach capacity increases
larger meals and go longer between snacks. The child can

Cardiovascular and respiratory system

The heart rate gradually decreases averaging
70-100 b/min
The blood pressure gradually increases
The lung become more mature, respiration

slow down.

Immune and lymphatic system

Hypertrophy of lymph tissue - tonsil and
adenoid are at their largest during this stage.
child has been exposed to many common
virus and bacteria leading to a much stronger immune response.

Reproductive system

Pre adolescence - there is a slow increase in hormone
production. Thearche may occur.

Gross motor development

Refine acquire skills

Runs, jumps, climbs, hops

Regular exercise

Fine motor development

use both hands independently

Draws a person with 10-20 parts

co-ordination continues to improve

Growth and development of school age

Definition

The school age typically refers to children between the ages of 6 to 12 years of age.

Physical and biological development

Vital signs

Temperature 36.5°C

Pulse 85 ± 10 beats/min

Respiration 20 ± 3 breath/min

Blood pressure $100/60 \pm 16/10$ mm of Hg.

Weight

Average weight is 18 - 40 Kgs

Expected weight = $\frac{(\text{Age in years} \times 7) - 5}{2}$

The weight increasing of 2 - 3 kg/year.

Height

Average height is 116 - 150 cms

Expected height = $\text{Age in years} \times 6 + 77$

The height increase of 5 cms/year.

Musculo skeletal system

Growth occurs primarily in the long bones

Muscle tissue increases in both length and thickness

Neurological system

Myelination continued insulation of nerve fibers

The neurological ability to filter out distractions improves

The brain and hand synchronize allows for complex task like playing a musical instrument, cursive writing

Safety and injury

Aspiration and choking
Positioning
Falls and drowning

accidents

why

Needs of toddlers

Routine and ritualism
Safe boundaries
Opportunities for autonomy
Sensory exploration

Role of parents

Safety officer
Architect of routine
Emotional coach
Language partner
Promoter of independence

Role of Nurse

Development screeners
Safety educator
Behavioral interpreter
Nutritional consultant
Primary immunizer

Intellectual development

Sensory motor stage (15-18 months)

Pre operational stage (24-30 months)

Moral development

Pre conventional morality

Language development

Receptive language

1 year $\rightarrow$ 6-8 words

2 years $\rightarrow$ 300 words

3 years $\rightarrow$ 900 words.

Play

parallel play

Nutrition

calories : 1300 - 1500 kcal/kg/day

protein : 1.7 to 2gms/kg/day

Fat : 4-5 gm/kg/day

Iron : 15-20 mg/day

calcium : 500mg/day

Fluid : 90-100 ml/kg/day

Common problems

Behavioral Problems

Temper tantrums

Negativism

Biting and hitting

Nutritional problems

physiologic anorexia

Food jags

Iron deficiency anaemia

Developmental
Problems

Health
Problems

Chemical
Problems

walks up and down stairs

30 months

stands on one foot alone momentarily
can throw a large ball overhand 4-5 feet
Rides a walker or pedal car
Jumps well in place off floor

fine motor development

15 months

Assume finger to thumb
use a neat pincer grasp to pick up small bits of foods
Drop a small object into narrow opening

18 months

Holds cup with both hand
Holds a cup grasp a spoon
Eats with spoon but spills
Making line with crayon
Begins turns pages
Remove simple garments

24 months

Builds tower of 6-7 cubes
pulls on own simple garments
Drinks well in one hand

30 months

Self feeding with occasional spilling
Buttons one large front button
unbuttons large front button

psycho social development

Autonomy vs shame and doubt

psycho sexual development

Anal stage

Spiritual development

instinctive projective faith

Gastro intestinal system

The stomach capacity accommodates larger, less frequent meals which matches the transition milk to solid table foods.

physiologic anorexia - This often looks like picky eating to parents but it's normal.

Respiratory system

The throat and upper respiratory tract remain small making them more susceptible to obstruction.

Immune system

The toddler begins to produce their own antibodies.

The maternal immunity passed at birth has

completely faded.

Integumentary system

The epidermis and dermis become more tightly bonded.

The skin becomes more effective at regulating temperature through shivering and sweating.

Gross motor development

15 months

Assume standing position

walks without support

creeps upstairs

18 months

walks quickly

Runs often walks

walks upstairs by holding one hand

24 months

Throws ball over hand without falling

Steady gait

Can walk heel-to-toe gait

Imitate well.

physical and biological development

vital signs

Temperature 36.5°C

pulse 90 - 130 beats/min

Respiration 26 - 28 breaths/min

Blood pressure 90/60 +/- 26/21 mmHg

weight and height

weight = 10 - 14 kg

expected weight = Age in years $\times 2 + 8$

At the age of 2.5 years the child attains 4 times than weight at birth.

Height = 83 - 95 cms

expected Height = Age in years $\times 6 + 77$ cms

dentition

At 2 years - 16 teeth

At 2.5 years - 20 teeth

Musculo skeletal system

The long bones continue to ossify.

The wide based toddle waddle eventually narrows as balance improves and the legs move from bow legged to straight alignment.

Neurological system

Myelination accelerates allowing for faster signal transmission

Brain volume is about 75% to 80% of its adult weight

pruning and synaptogenesis allow for complex language and problem solving.

Myelination of the spinal cord allows voluntary control of the anal and urethral sphincters

Master Plan on Growth and Development of Toddlers and School age

Introduction

Growth and development starts before the baby is born from the conception in mother's womb. The period extends throughout the life cycle.

Growth

It is the process of physical maturation results in an increase in size of the body & various organs. It occurs by multiplication of cells and an increase in intracellular substance. It is a quantitative changes of the body which can be measured in inches/centimeters and pounds/kilograms.

Development

It is the process of functional and physiological maturation of the individual. It is progressive increase in skill and capacity to function. It related to maturation and myelination of the nervous system.

Maturation

It is an increase in competence and changes in behavior and ability to function at higher level depending upon the genetic inheritance.

Growth and development of Toddlers

Definition of Toddlers

Toddler is a child between the ages of 12-36 months or 1 to 3 years old. The term derived from the verb to ~~do~~ "toddle", referring to the unsteady, wide based and pre cursive way a child walks as they transition from crawling to independent movement.